

Spark daemon pattern: secondary review package (r4.5)

- **For:** secondary reviewers (people or models). This document stands alone: every file it names is in `spark-daemon-pattern/` on branch `claude/clever-bardeen-qqkxez`, for reviewers who can read the repository.
- **State at r4.5:**
 - contract 3.1.0;
 - 210 of 210 self-tests pass;
 - the four reference daemons pass all 19 battery checks (DB-01 to DB-18 and the new DB-20).
- **What to review first:** Part 1, the conformance battery and the test list, is the main request. Part 2 records the adjudication it builds on, including the two defects fixed in this revision (HF-34 and HF-35). Part 3 reviews the SQLite three-ledger recommendations.

How to mark: for each check, planned check, planned test or decision, give AGREE, CHANGE or REJECT. Say which evidence type you are relying on: reproduced, measured, traced or opinion. Part 1 §6 lists the questions this review most needs answered.

Part 1: Conformance battery review and test list

- **Status:** FOR SECONDARY REVIEW. It describes the battery and self-tests as they are at r4.5 (contract 3.1.0), what each check cannot catch, the gaps, and the planned checks and tests that the adjudication (`ADJUDICATION-R4.5.md`) calls for. Every number here was produced on this revision.
- **Revision:** r4.5, 2026-09-30, by Claude.
- **Evidence at this revision:**
 - 210 of 210 self-tests pass (Python 3.12, `-W error::ResourceWarning`).
 - The four reference daemons pass all 19 battery checks: 18 original plus DB-20.
 - The contract check is up to date (3.1.0).
 - The workspace has Landlock ABI 7, `strace` and `systemd-analyze`; it has no `systemd` PID 1, and its `cgroup` is v1.

Two kinds of evidence, kept apart on purpose:

- **The battery** judges *one daemon*. It runs that daemon's real code, in child processes, in a disposable workspace, and its PASS is the only admissible evidence for activating that daemon.
- **The self-tests** judge *the skeleton*: properties every daemon inherits, such as the ledger format, the blind clock, purity and the unit generator.

A property belongs in the battery when a daemon's own code or manifest can break it. Otherwise it belongs in the self-tests. Reviewers should challenge any item they think sits on the wrong side of that line.

1. Code map

What each module does, its size in lines, and where the adjudicated work will land.

Module	Lines	Responsibility	Adjudicated change that lands here
<code>manifest.py</code>	432	Closed-schema manifest and validation; <code>blind_limit_seconds</code> range	Contract 4.0.0: pacing (PD-85), consequence level (PD-01.2), schema-3 migration hint
<code>runtime.py</code>	576	Start-up order, the cycle loop, the blind clock, heartbeats, stop	r4.5: boot-time basis for inherited blindness (HF-34). Contract 4.0.0: pacing
<code>context.py</code>	183	The only I/O a daemon may perform (<code>read_text</code> , <code>list_dir</code> , <code>stat</code> , <code>run</code> , <code>git</code> , <code>unsettled</code>)	PD-83: truncation always raises TooLarge
<code>proc.py</code>	217	Bounded, allowlisted subprocesses; the Git hardening profile	—
<code>guard.py</code>	258	Path policy, output-directory checks, the audit hook	—
<code>landlock.py</code>	246	Kernel confinement (the only <code>ctypes</code> besides probes)	—
<code>purity.py</code>	215	Static purity check before import	—
<code>ledger.py</code>	256	Append-only JCS hash chain; recovery; quarantine	An independent verifier is planned <i>outside</i> this package (B-6)
<code>canonical.py</code>	118	Canonical JSON and hashing	Shared rules for any other ledger format (see <code>SQLITE-LEDGER-REVIEW.md</code>)
<code>render.py</code>	176	Structure-safe digest	—
<code>unitgen.py</code>	250	The <code>systemd</code> unit and its lint	The exit-code table (N-12, PD-63's split)
<code>notify.py</code>	23	<code>sd_notify</code>	—

Module	Lines	Responsibility	Adjudicated change that lands here
contract.py	374	The machine-readable author contract	3.1.0 now; 4.0.0 batched
battery.py	800	DB-01 to DB-18 and DB-20	r4.5: DB-20. Planned: DB-19, DB-21, DB-22 (\$4)
probes.py	266	Child-process probes the battery uses	—
handoff.py, scaffold.py, cli.py, paths.py, __init__.py	684	Candidate envelopes, validate and precheck, command line, constants	Exit codes: C-1 and PD-63's split, in contract 4.0.0

2. The checks, one by one

"Cannot catch" is the list a secondary reviewer should attack. **Independent** says whether the check uses code other than the skeleton code it judges.

ID	What it checks, and how	Cannot catch	Independent?	Environment
DB-01	The manifest validates against the closed schema	Limits that are valid alone but wrong together (U-8). Only a few cross-checks exist today (planned DB-21)	Uses the skeleton's validator	Any
DB-02	The static purity check of daemon.py	Routes the syntactic rules do not know (residual risk 2 in HARDENING.md). Landlock is the backstop, which DB-17 proves	Skeleton's checker	Any
DB-03	Confinement: 5 cycles, then a snapshot comparison of the workspace HOME and the daemon folder; no audit-hook events	Writes outside the snapshot roots (Landlock and DB-17 cover these); behaviour that needs inputs the fixture lacks	Snapshot is independent; the audit count comes from the skeleton	Any
DB-04	Ledger integrity and provenance after a normal run: verifies, starts with DAEMON_START, ends with DAEMON_STOP, only declared types, zero DAEMON_ERROR, digests match	A daemon that never accepts a cycle (now DB-20, HF-35). Verifier defects, because it uses the skeleton's own verifier (B-6)	No	Any
DB-05	Crash and restart: 8 SIGKILLs (3 with --quick) at seeded random points; the chain verifies; every restart is flagged unclean; quarantine files match records	Power loss: SIGKILL leaves the page cache intact, so an fsync that never happened cannot be seen	Skeleton's verifier	Any
DB-06	Torn tails (garbage; a complete record without its newline) are quarantined byte for byte and recorded	Other tear shapes (a fuzz corpus is a candidate addition)	Skeleton's verifier	Any
DB-07	One flipped byte in a middle record: the daemon refuses to start (65) and changes nothing	Only one corruption shape	Skeleton's verifier	Any
DB-08	Injected fsync EI0: exit 70 at once, and a restart recovers	—	strace fault injection	Needs strace , otherwise UNKNOWN, so INCOMPLETE
DB-09	Injected ENOSPC on write: exit 70, and a restart recovers	A real full disk with partial writes	strace	Needs strace
DB-10	The digest stays structure-safe under hostile values (seeded probe)	Daemons without a digest (N/A)	Probe	Any
DB-11	The notify protocol against a stand-in NOTIFY_SOCKET: READY after DAEMON_START, watchdog pings, STOPPING	Real systemd behaviour (it has never run under PID 1)	Stand-in socket	Any
DB-12	A second instance is refused (73); SIGTERM gives DAEMON_STOP and exit 0	—	—	Any
DB-13	The audit hook blocks the fixed list of forbidden operations for this manifest	Operations not on the list	Probe, in-process	Any
DB-14	Resource budget: projected CPU per cycle, and peak memory as max(wait4 ru_maxrss, DAEMON_STOP RUSAGE_SELF) over 21 cycles	Memory held by a daemon while its child runs: measured 38 % low (evidence/r4.4/). Worst-case inputs (fixture data is small, PD-71)	Partly	Any. To be replaced by DB-19 (PD-76)
DB-15	The unit carries every required directive; systemd-analyze security --offline exposure is under the threshold; start-up syscalls are allowed by the unit's seccomp filter	The unit's behaviour under a real service manager	systemd-analyze	Needs systemd-analyze , otherwise UNKNOWN
DB-16	Verification is read-only and repeatable: two runs agree, bytes and mtime unchanged	"Two runs agree" runs the same code twice (B-6)	No	Any

ID	What it checks, and how	Cannot catch	Independent?	Environment
DB-17	With the audit hook in record-only mode, the kernel alone blocks a forbidden write, a denied read and a TCP connect	Kernels without Landlock (the check becomes UNKNOWN; the runtime refuses to start there outside test mode)	Kernel	Landlock ABI dependent
DB-18	DAEMON_START.landlock is present and matches this host's ABI and the manifest's gaps	Another host's ABI: a PASS here says nothing about the DGX's ABI (B-5)	—	Host specific
DB-20 (r4.5)	The daemon observes: 12 cycles against a blind limit of five test intervals (1.0 s); accepted cycles appear in the heartbeats; no SENSE_BLIND	A daemon that observes the fixture but not production inputs (PD-71). Slow first cycles longer than 1.0 s would be reported as blind (none of the reference daemons comes close)	Uses the runtime's own counters	Any

Verdict rule:

- PASS only when nothing FAILS and nothing is UNKNOWN;
- FAIL if anything FAILS;
- INCOMPLETE otherwise.

Exit codes are PASS 0, INCOMPLETE 3, FAIL 1. They change to 0/3/4 in contract 4.0.0 (C-1). The report binds the manifest digest, the code digest and the contract identity.

3. Gaps, in order of risk

#	Gap	Consequence	Plan
G-1	Memory is read per process (DB-14)	A daemon can pass while its cgroup exceeds MemoryMax= (38 % under-read, measured)	DB-19 (PD-76), with a separate cgroup helper
G-2	The ledger checks are not independent (DB-04, DB-16)	A verifier defect passes its own output	DB-22: a verifier that does not import spark_daemon, cross-checked with a second JCS implementation
G-3	Limits are not checked against each other in the battery	HF-28, HF-30, HF-31 and HF-32 were all limit interactions	DB-21: the budget table (PD-69), including time windows (PD-86)
G-4	The fixtures are small	Worst-case behaviour (a huge repository, a full inbox) is never exercised	Worst-case fixtures per observation type, plus a scaling series (PD-71). The first such fixture is dir-watch over its cap (HF-33)
G-5	PASS is host-specific but does not say so	A PASS on this workspace is read as a PASS on the DGX	Bind a host fingerprint (kernel, Landlock ABI, cgroup version, tool versions) into the report (B-2); host-caused N/A counts as an owed environment
G-6	Never run under systemd PID 1	DB-11 and DB-15 simulate	First DGX step: install one reference daemon and confirm DAEMON_START.landlock.status == "enforced"
G-7	Two checks need tools (strace, systemd-analyze)	Without them the verdict is INCOMPLETE, which is correct, but CI must decide what that means	The adjudicated CI rule (INCOMPLETE only with a stated environment reason)
G-8	Power-loss durability is not provable by SIGKILL	Missing fsyncs would pass DB-05	DB-08 covers fsync errors. A write-order trace of fsync placement is a candidate check

4. Planned checks

New IDs are never edited in place (PD-78).

ID	Check	From	When
DB-19	A direct cgroup peak (v2 memory.peak, or v1 memory.max_usage_in_bytes against a v1 limit) for a fresh cgroup holding the daemon and every child from its first instruction. Replaces DB-14's memory half; DB-14 is retired and its CPU half carried over	PD-76 (owner), DECISION-BRIEF.md §2.4	Next
DB-21	A budget table: worst-case cycle against the watchdog (at most half), stop timeout against one watchdog period, blind limit against intervals, and retry bounds over time	PD-69, PD-86, U-8	With Phase 1's budget report
DB-22	Independent ledger verification: a separate verifier and a second JCS implementation agree with the skeleton's	B-6, U-7	Phase 2
DB-23	Worst-case fixtures per observation type: over-capacity input fails the cycle and never records a partial result	PD-71, HF-33, PD-83	With DB-19
DB-2x	The LTC provider-neutral suite for each declared connection	PD-82, LI-11	With the first request channel

5. Planned self-tests (skeleton properties)

ID	Test	From
T-EX1	One exit-code table drives the runtime constants, RestartPreventExitStatus and the notifier mapping; the test fails if any of them drifts	PD-63 split, N-12
T-EX2	EXIT_POLICY and EXIT_SENSE_BLIND differ, and both are never restarted	PD-63 split
T-TR1	ctx.run, ctx.git and ctx.list_dir raise TooLarge one byte or one entry past the cap, and succeed exactly at it	PD-83 (modified)
T-PC1	pacing: BOUNDED_JITTER stays within its bound; DETERMINISTIC_SPREAD gives the same phase on restart; FIXED_PHASE injects nothing; watchdog pings continue through any offset	PD-85, N-11
T-ST1	The staleness checker flags a ledger that stops changing within T, using its own monotonic clock, and is unaffected by a wall-clock step	Phase 1 (modified), N-5
T-CG1	The cgroup helper places the child before exec, cleans up on every exit path, and records the cgroup version	DB-19, N-4
T-N1 to T-N14	One test per seam when its decision is built: supervisor/worker (N-1), register binding (N-2), declared connection and LTC bounds (N-3), cgroup (N-4), staleness (N-5), inbound acknowledgement never carries commands (N-6), gate epoch fencing (N-7), witness independence (N-8), recorder anchoring (N-9), pacing and watchdog (N-11), exit table (N-12), schema-3 migration (N-13), survival mode never read as authority (N-14). N-10 is removed by PD-83's modification	REVIEW-PACKAGE-COMBINED.md, reviewer focus

6. Questions for the secondary reviewer

1. Is any check on the wrong side of the battery/self-test line (§ introduction)?
2. DB-20 uses a 1.0 s blind limit over 12 cycles. Is there a legitimate daemon whose first accepted cycle takes longer in the battery workspace, and should the limit scale with the manifest's step_timeout_seconds?
3. Is `max(ru_maxrss)` ever the *right* basis once DB-19 exists, for example for a daemon with no children?
4. Should DB-05 add a power-loss simulation (for example a `dm-flakey` or `fsync-trace` check), or is DB-08 plus code review enough?
5. For G-5: which host facts must a PASS bind before it can be used on another machine?
6. Which of the 210 self-tests below would you delete as redundant, and which property has no test at all?

7. The self-test list

Generated from the test files at r4.5 (`evidence/r4.5/testlist.py`). "Cites" lists the defect (HF), decision (PD), battery check (DB) and rule identifiers each test names in its body or docstring.

tests/test_blind.py (29 tests)

Blind period (r3.5): `ctx.unsettled()`, `blind_limit_seconds` and `SENSE_BLIND`, and the unit

Class	Test	What it pins	Cites
BlindPeriodTests	<code>test_unsettled_is_neither_an_event_nor_an_error</code>	Unsettled is neither an event nor an error.	
BlindPeriodTests	<code>test_always_unsettled_stops_at_the_limit</code>	Always unsettled stops at the limit.	
BlindPeriodTests	<code>test_failed_cycles_count_towards_the_limit</code>	r2's silent-outage shape: a daemon failing every cycle kept pinging the watchdog forever.	
BlindPeriodTests	<code>test_an_accepted_cycle_resets_the_clock</code>	Busy for about twice the limit in total, but never for a whole limit at once.	
BlindPeriodTests	<code>test_blindness_survives_a_restart_loop</code>	HF-32: killed and restarted every 1.0 s against a 1.5 s limit. On r3.9 no run ever reached the limit; now the	HF-32

Class	Test	What it pins	Cites
		second run inherits the first run's blindness.	
BlindPeriodTests	test_a_backward_clock_step_does_not_hide_blindness	On r4.4 every restart after a one-hour backward step inherited 0 ms, and six kill-restarts (6 s blind against a 1.5 s limit) never reached SENSE_BLIND (evidence/r4.5/).	HF-34
BlindPeriodTests	test_start_heartbeat_and_stop_carry_the_boot_stamp	Start heartbeat and stop carry the boot stamp.	HF-34
BlindPeriodTests	test_across_a_reboot_the_wall_clock_is_used_and_a_backward_one_assumes_the_worst	Across a reboot the wall clock is used and a backward one assumes the worst.	HF-34
BlindPeriodTests	test_an_event_after_the_last_stamp_is_anchored_at_that_stamp	A daemon event carries no boot time; the preceding stamp bounds it from below (safe side).	HF-34
BlindPeriodTests	test_a_clean_stop_does_not_reset_blindness	A clean stop does not reset blindness.	HF-32
BlindPeriodTests	test_a_restart_past_the_limit_gets_one_reacquisition_cycle	A restart past the limit gets one reacquisition cycle.	HF-32
BlindPeriodTests	test_heartbeats_tell_quiet_from_dead	A healthy daemon that sees no change still leaves evidence every limit/2.	HF-32
BlindPeriodTests	test_a_malformed_reason_is_a_sense_failure	A malformed reason is a sense failure.	
BlindPeriodTests	test_the_override_is_ignored_outside_test_mode	Outside test mode the manifest's 600 s applies, and the interval is the manifest's	
BlindLimitManifestTests	test_required_with_no_default	Required with no default.	
BlindLimitManifestTests	test_bounded_both_ways	Bounded both ways.	
BlindLimitManifestTests	test_at_least_three_intervals	At least three intervals.	
BlindLimitManifestTests	test_version_1_manifest_gets_a_migration_hint	Version 1 manifest gets a migration hint.	
GitWatchUnsettledTests	test_moving_refs_are_unsettled	Moving refs are unsettled.	
GitWatchUnsettledTests	test_a_changing_worktree_is_unsettled	A changing worktree is	

Class	Test	What it pins	Cites
		unsettled.	
GitWatchUnsettledTests	test_output_over_the_cap_fails_the_cycle	Before r3.5 this was accepted as "dirty: null" every cycle, so it never reached the limit.	
GitWatchUnsettledTests	test_a_still_repository_is_observed	A still repository is observed.	
DirWatchCapacityTests	test_a_folder_over_the_cap_fails_the_cycle	A folder over the cap fails the cycle.	HF-33
DirWatchCapacityTests	test_adding_a_file_over_the_cap_never_reports_a_removal	Adding a file over the cap never reports a removal.	HF-33
DirWatchCapacityTests	test_at_the_cap_the_inventory_is_complete	At the cap the inventory is complete.	HF-33
NoRestartTests	test_fail_closed_exits_are_never_restarted	Fail closed exits are never restarted.	
NoRestartTests	test_a_stop_outlasts_one_watchdog_period	r3.9 (HF-31): a stop is honoured between cycles; a fixed 30 s SIGKILLED slow cycles.	HF-31
NoRestartTests	test_git_watch_worst_case_sense_fits_half_the_watchdog	r3.9 (HF-30): six Git calls per worktree cycle since r3.5, each up to step_timeout.	HF-30
NoRestartTests	test_uncertain_commit_stays_restartable	Uncertain commit stays restartable.	

tests/test_canonical.py (9 tests)

Canonical JSON: golden vector, refusals, determinism, strict parsing.

Class	Test	What it pins	Cites
CanonicalTests	test_golden_vector	Golden vector.	
CanonicalTests	test_genesis_is_64_zeros	Genesis is 64 zeros.	
CanonicalTests	test_key_order_matches_rfc8785_not_code_point_order	Key order matches rfc8785 not code point order.	
CanonicalTests	test_key_order_nested	The reordering must apply at every depth, not only the top level.	
CanonicalTests	test_allowed_conversions	Allowed conversions.	
CanonicalTests	test_refusals	Refusals.	
CanonicalTests	test_lone_surrogate_refused	Lone surrogate refused.	
CanonicalTests	test_independent_of_hash_seed	Independent of hash seed.	
CanonicalTests	test_strict_loads	Strict loads.	

tests/test_guard.py (8 tests)

Guard: output-directory checks, inventory, path policy, and the audit hook in a child process.

Class	Test	What it pins	Cites
OutputDirTests	test_creates_0700_tree	Creates 0700 tree.	

Class	Test	What it pins	Cites
OutputDirTests	test_refuses_loose_permissions	Refuses loose permissions.	
OutputDirTests	test_refuses_symlink	Refuses symlink.	
OutputDirTests	test_stray_temp_files_removed	Stray temp files removed.	
OutputDirTests	test_inventory_reports_foreign_files	Inventory reports foreign files.	
AuditHookTests	test_enforce_mode_blocks_every_forbidden_operation	Enforce mode blocks every forbidden operation.	
AuditHookTests	test_record_mode_reports_without_blocking	Record mode reports without blocking.	
AuditHookTests	test_policy_readable	Policy readable.	

tests/test_handoff.py (31 tests)

The handoff layer: the published contract, the manifest JSON Schema, candidate envelopes,

Class	Test	What it pins	Cites
PublishedContractTests	test_committed_contract_matches_the_running_rules	Committed contract matches the running rules.	
PublishedContractTests	test_committed_schema_matches_the_running_validator	Committed schema matches the running validator.	
PublishedContractTests	test_contract_version_is_pinned_to_its_sha	Changing any enforced rule changes contract_sha256. That must come with a new CONTRACT_VERSION and a new line in contract/versions.json, never a silent edit.	
PublishedContractTests	test_contract_is_identical_on_every_installed_python	Contract is identical on every installed python.	
PublishedContractTests	test_contract_names_every_ctx_method_and_battery_check	Contract names every ctx method and battery check.	
PublishedContractTests	test_describe_cli	Describe cli.	
SchemaAgreementTests	test_schema_is_valid_draft_2020_12	Schema is valid draft 2020 12.	
SchemaAgreementTests	test_every_shipped_manifest_passes_both	Every shipped manifest passes both.	
SchemaAgreementTests	test_mutations_are_refused_by_both	Mutations are refused by both.	
SchemaAgreementTests	test_cross_field_rules_are_manifest_py_only_and_declared	Cross field rules are manifest py only and declared.	
EnvelopeTests	test_scaffold_envelope_is_exact	Scaffold envelope is exact.	
EnvelopeTests	test_a_candidate_cannot_certify_itself	A candidate cannot certify itself.	
EnvelopeTests	test_files_changed_after_sealing	Files changed after sealing.	
EnvelopeTests	test_other_major_version_is_incompatible	Other major version is incompatible.	
EnvelopeTests	test_same_version_other_rules_is_a_mismatch	Same version other rules is a mismatch.	
EnvelopeTests	test_same_major_other_minor_is_a_warning	Same major other minor is a warning.	
EnvelopeTests	test_bad_producer_kind	Bad producer kind.	
EnvelopeTests	test_envelope_cli_round_trip	Envelope cli round trip.	
ValidateJsonTests	test_structured_diagnostics	Structured diagnostics.	

Class	Test	What it pins	Cites
ValidateJsonTests	test_manifest_diagnostics_carry_the_field	Manifest diagnostics carry the field.	
ValidateJsonTests	test_text_mode_is_unchanged_for_people	Text mode is unchanged for people.	
PrecheckTests	test_every_reference_daemon_prechecks_ok	Every reference daemon prechecks ok.	
PrecheckTests	test_impure_daemon_fails_fast_and_skips_the_run	Impure daemon fails fast and skips the run.	
PrecheckTests	test_policy_violation_is_caught_by_the_short_run	Policy violation is caught by the short run.	
PrecheckTests	test_precheck_never_says_pass	Precheck never says pass.	
ScaffoldTests	test_refuses_a_non_empty_folder_and_bad_names	Refuses a non empty folder and bad names.	
ScaffoldTests	test_scaffold_passes_validate_and_user_unit_variant	Scaffold passes validate and user unit variant.	
SymlinkStatTests	test_planted_symlink_is_a_link_not_a_violation	Planted symlink is a link not a violation.	HF-17
SymlinkStatTests	test_following_reads_still_judge_the_target	Following reads still judge the target.	HF-17
BatteryCatchesAlwaysFailingDaemonsTests	test_a_daemon_that_errors_every_cycle_fails_db04	r2 passed the raiser fixture: its sense() fails every cycle in the battery's workspace (no ~/data/mode.txt), yet the process exits 0 with a valid chain.	DB-04, HF-16
BatteryCatchesAlwaysFailingDaemonsTests	test_a_daemon_that_never_obeys_fails_db20	r4.5 (HF-35): every cycle unsettled means no DAEMON_ERROR and no event; with its digest disabled, r4.4's battery reported RESULT: PASS for it.	DB-04, DB-20, HF-35

tests/test_hardening.py (40 tests)

Regression tests for the r3 hardening: every escape or crash found in the r2 review has a

Class	Test	What it pins	Cites
ManifestHardeningTests	test_interpreter_families_and_command_runners_are_refused	Interpreter families and command runners are refused.	HF-11
ManifestHardeningTests	test_harmless_commands_still_allowed	Harmless commands still allowed.	
ManifestHardeningTests	test_dot_segments_and_trailing_slashes_refused	Dot segments and trailing slashes refused.	HF-14
ManifestHardeningTests	test_output_dir_must_not_contain_a_protected_tree	r2 accepted "~": Landlock and ReadWritePaths= would then grant write access to the whole home directory, ~/.ssh and ~/spark-core included.	HF-12
ManifestHardeningTests	test_output_dir_inside_the_manifests_own_deny_is_refused	Output dir inside the manifests own deny is refused.	
ManifestHardeningTests	test_trailing_newlines_are_refused	r2 used re.match with "\$", which also matches before a final newline: a name of "x\n" split the generated unit's Description= line in two.	HF-13
ManifestHardeningTests	test_to_dict_is_bounded_and_strict	To dict is bounded and strict.	
GitHardeningTests	test_refusals	Refusals.	
GitHardeningTests	test_allowed	Allowed.	

Class	Test	What it pins	Cites
GitHardeningTests	test_diff_subcommands_get_no_ext_diff_and_no_textconv	Diff subcommands get no ext diff and no textconv.	HF-08, HF-19, PD-25
GitHistoryIntegrityTests	test_repository_gpg_program_never_runs	Repository gpg program never runs.	
GitHistoryIntegrityTests	test_replace_refs_do_not_forge_history	Replace refs do not forge history.	
GitHistoryIntegrityTests	test_grafts_do_not_rewrite_ancestry	Grafts do not rewrite ancestry.	
GitInjectionRuntimeTests	test_alias_injection_through_ctx_git_is_refused	Alias injection through ctx git is refused.	
GitInjectionRuntimeTests	test_git_through_ctx_run_is_refused	Git through ctx run is refused.	HF-02
GitInjectionRuntimeTests	test_read_only_git_still_works	Read only git still works.	
PurityHardeningTests	test_frame_escape_to_real_builtins	Frame escape to real builtins.	HF-04
PurityHardeningTests	test_traceback_frame_escape	Traceback frame escape.	HF-04
PurityHardeningTests	test_private_attribute	Private attribute.	HF-03
PurityHardeningTests	test_aliasing_a_forbidden_builtin	Aliasing a forbidden builtin.	HF-05
PurityHardeningTests	test_dynamic_attribute_helpers	Dynamic attribute helpers.	HF-06
PurityHardeningTests	test_format_string_reaching_a_dunder	Format string reaching a dunder.	HF-06
PurityHardeningTests	test_exit_and_quit	Exit and quit.	
PurityHardeningTests	test_import_time_calls	Import time calls.	
PurityHardeningTests	test_entry_points_cannot_be_reassigned_or_redefined	Entry points cannot be reassigned or redefined.	
PurityHardeningTests	test_allowed_idioms_still_pass	Allowed idioms still pass.	
PurityHardeningTests	test_every_shipped_daemon_is_still_pure	Every shipped daemon is still pure.	
RuntimeHardeningTests	test_system_exit_in_sense_is_recorded_not_a_silent_exit	r2: raise SystemExit(0) ended the process with exit 0 and no DAEMON_STOP, so systemd's Restart=on-failure would never have restarted it.	HF-09
RuntimeHardeningTests	test_undeclared_command_request_fails_closed_even_when_swallowed	Undeclared command request fails closed even when swallowed.	HF-10
RuntimeHardeningTests	test_rejected_duplicate_launch_touches_nothing	r3.2 (WBS 3.1 §4.2): the second instance cleaned tmp/ before taking the lock and deleted the running instance's in-flight files (for git-watch, its index copy).	HF-25
RuntimeHardeningTests	test_torn_first_write_still_starts_the_ledger_with_daemon_start	r3.2 (WBS 3.0 r4's pre-baseline correction): a crash during the first-ever append left bytes but no complete record; the next start wrote LEDGER_TAIL_QUARANTINED as seq 1, a ledger the battery's own DB-04 then ...	DB-04, HF-26
RuntimeHardeningTests	test_policy_is_immutable	Policy is immutable.	HF-03
CpuAccountingTests	test_cpu_includes_commands_the_daemon_ran	Cpu includes commands the daemon ran.	HF-18
UnitHardeningTests	test_unit_allows_the_landlock_syscalls	Unit allows the landlock syscalls.	HF-07
UnitHardeningTests	test_resolve_syscall_filter	Resolve syscall filter.	HF-07
UnitHardeningTests	test_unsafe_paths_are_refused	Unsafe paths are refused.	HF-22
RenderLabelTests	test_label_with_trailing_newline_is_refused	Label with trailing newline is refused.	HF-13

Class	Test	What it pins	Cites
RenderEfficiencyTests	test_binary_search_matches_the_linear_scan	Binary search matches the linear scan.	HF-21
CliRobustnessTests	test_verify_with_a_bad_manifest_reports_instead_of_crashing	Verify with a bad manifest reports instead of crashing.	
CliRobustnessTests	test_battery_with_invalid_json_reports_instead_of_crashing	Battery with invalid json reports instead of crashing.	

tests/test_landlock.py (12 tests)

Landlock (AP-01): ABI detection, the pure helpers, and real enforcement in forked

Class	Test	What it pins	Cites
AbiTests	test_abi_version_is_an_int	Abi version is an int.	
AbiTests	test_this_workspace_kernel_supports_landlock	Documents the environment this suite runs in; the DGX (aarch64, a different kernel) must be checked separately per ADJUDICATION-AP.md.	
PureHelperTests	test_fs_access_mask_grows_with_abi	Fs access mask grows with abi.	
PureHelperTests	test_fs_access_for_file_drops_directory_only_rights	Fs access for file drops directory only rights.	
PureHelperTests	test_fs_access_for_directory_keeps_every_requested_right	Fs access for directory keeps every requested right.	
PureHelperTests	test_gaps_finds_a_denied_path_inside_a_granted_read	Gaps finds a denied path inside a granted read.	
PureHelperTests	test_gaps_empty_when_reads_and_denies_do_not_overlap	Gaps empty when reads and denies do not overlap.	
SupervisorDomainTestModeTests	test_unavailable_landlock_is_non_fatal_in_test_mode	Unavailable landlock is non fatal in test mode.	
SupervisorDomainTestModeTests	test_unavailable_landlock_refuses_outside_test_mode	Unavailable landlock refuses outside test mode.	
EnforcementTests	test_supervisor_domain_blocks_denied_reads_and_outside_writes	Supervisor domain blocks denied reads and outside writes.	
EnforcementTests	test_nested_child_domain_can_only_narrow_never_widen	Nested child domain can only narrow never widen.	AP-04
EnforcementTests	test_output_directory_gets_every_right_the_abi_handles	Output directory gets every right the abi handles.	

tests/test_layers.py (4 tests)

Framework / services boundary (r3.4, PD-53; a Python reading of the Asterinas framkernel

Class	Test	What it pins	Cites
LayerBoundaryTests	test_service_modules_import_nothing_that_reaches_the_os	Service modules import nothing that reaches the os.	
LayerBoundaryTests	test_service_modules_never_call_open	Service modules never call open.	
LayerBoundaryTests	test_ctypes_only_where_declared	Ctypes only where declared.	
LayerBoundaryTests	test_exactly_one_truncate_site	r4's audit rule: truncation destroys bytes, so it happens in one reviewed place (ledger.recover, after the torn tail is copied to quarantine).	

tests/test_ledger.py (16 tests)

Ledger: append and verify, every corruption category, torn tails, uncertain commits.

Class	Test	What it pins	Cites
LedgerTests	test_append_and_verify	Append and verify.	
LedgerTests	test_flipped_byte	Flipped byte.	

Class	Test	What it pins	Cites
LedgerTests	test_swapped_lines	Swapped lines.	
LedgerTests	test_missing_line	Missing line.	
LedgerTests	test_non_canonical_line	Non canonical line.	
LedgerTests	test_foreign_daemon	Foreign daemon.	
LedgerTests	test_newer_schema_fails_explicitly	Newer schema fails explicitly.	
LedgerTests	test_overlong_line	Overlong line.	
LedgerTests	test_corrupt_last_complete_line_is_not_a_torn_tail	Corrupt last complete line is not a torn tail.	
LedgerTests	test_torn_tail_garbage_is_quarantined	Torn tail garbage is quarantined.	
LedgerTests	test_complete_record_without_newline_is_quarantined	Complete record without newline is quarantined.	
LedgerTests	test_two_torn_tails_never_overwrite	Two torn tails never overwrite.	
LedgerTests	test_prepare_refuses_before_writing	Prepare refuses before writing.	
LedgerTests	test_fsync_failure_is_uncertain	Fsync failure is uncertain.	
LedgerTests	test_short_write_is_uncertain	Short write is uncertain.	
LedgerTests	test_streaming_pass_memory_does_not_depend_on_length	Streaming pass memory does not depend on length.	

tests/test_manifest.py (15 tests)

Manifest: the example validates; every refusal category is enforced.

Class	Test	What it pins	Cites
ManifestTests	test_example_is_valid	Example is valid.	
ManifestTests	test_hash_ignores_key_order_and_whitespace	Hash ignores key order and whitespace.	
ManifestTests	test_unknown_keys_refused_at_every_level	Unknown keys refused at every level.	
ManifestTests	test_missing_key	Missing key.	
ManifestTests	test_acting_daemons_refused	Acting daemons refused.	
ManifestTests	test_network_refused	Network refused.	
ManifestTests	test_forbidden_commands	Forbidden commands.	
ManifestTests	test_reads_inside_denied_paths	Reads inside denied paths.	
ManifestTests	test_output_dir_placement	Output dir placement.	
ManifestTests	test_reserved_event_types	Reserved event types.	
ManifestTests	test_step_timeout_bounded_by_watchdog	Step timeout bounded by watchdog.	
ManifestTests	test_system_unit_needs_non_root_user	System unit needs non root user.	
ManifestTests	test_paths_limited_to_safe_characters	Paths limited to safe characters.	
ManifestTests	test_purpose_cannot_carry_systemd_specifiers	Purpose cannot carry systemd specifiers.	
ManifestTests	test_floats_refused	Floats refused.	

tests/test_misc.py (12 tests)

Unit generator, purity check, battery and source hygiene.

Class	Test	What it pins	Cites
UnitgenTests	test_example_unit_carries_every_required_directive	Example unit carries every required directive.	
UnitgenTests	test_lint_detects_a_removed_directive	Lint detects a removed directive.	
UnitgenTests	test_user_units_carry_a_warning	User units carry a warning.	
UnitgenTests	test_install_plan_is_text_only	Install plan is text only.	
PurityTests	test_good_module	Good module.	
PurityTests	test_example_is_pure	Example is pure.	
PurityTests	test_each_forbidden_construct	Each forbidden construct.	
PurityTests	test_required_functions_and_arity	Required functions and arity.	

Class	Test	What it pins	Cites
BatteryTests	test_example_passes_or_is_incomplete_never_fails	Example passes or is incomplete never fails.	DB-18, DB-19, DB-20, HF-35, PD-76
BatteryTests	test_impure_daemon_fails	Impure daemon fails.	DB-02, DB-03
RuntimeStructureTests	test_gc_collect_runs_once_per_cycle_not_inside_the_sleep_loop	Structural guard for the multi-daemon GC-forcing addition (r2): gc.collect() must be called exactly once per cycle of the outer loop, and not inside the inner sleep loop (which would call it many times per cycle, ...	
SourceHygieneTests	test_no_raw_invisible_characters_in_source	Trojan-source guard: bidi, zero-width and control characters appear only as escapes.	

tests/test_proc.py (6 tests)

Hardened subprocess: allowlist, scrubbed environment, bounds, timeout, and the F4 index-copy fix.

Class	Test	What it pins	Cites
ProcTests	test_only_allowlisted_commands_run	Only allowlisted commands run.	
ProcTests	test_output_is_bounded	Output is bounded.	
ProcTests	test_timeout_kills_the_group	Timeout kills the group.	
ProcTests	test_git_ignores_user_configuration	Script-board F5: color.ui=always and diff.external in the user's config must not leak.	F5
ProcTests	test_inherited_git_selection_variables_cannot_redirect_git	r4.0 (hardening review, item 2): GIT_DIR, GIT_WORK_TREE and GIT_OBJECT_DIRECTORY in the daemon's own environment must not redirect a call to a decoy repository. Every child starts from SAFE_ENV (an allowlist), so ...	
ProcTests	test_index_copy_leaves_the_real_index_untouched	Script-board F4: porcelain git diff rewrites .git/index; the index copy does not.	F4

tests/test_render.py (10 tests)

Rendering: hostile values stay inside their blocks; structure never changes; bounds hold.

Class	Test	What it pins	Cites
RenderTests	test_hostile_corpus_is_inert	Hostile corpus is inert.	
RenderTests	test_structure_invariance_property	Structure invariance property.	
RenderTests	test_escapes_are_visible_and_unambiguous	Escapes are visible and unambiguous.	
RenderTests	test_multiline_values_are_prefixed	Multiline values are prefixed.	
RenderTests	test_fence_longer_than_any_backtick_run	Fence longer than any backtick run.	
RenderTests	test_labels_are_validated	Labels are validated.	
RenderTests	test_values_are_type_checked	Values are type checked.	
RenderTests	test_size_bound_truncates_with_marker	Size bound truncates with marker.	
RenderTests	test_stamp_is_validated	Stamp is validated.	
RenderTests	test_deterministic	Deterministic.	

tests/test_runtime.py (18 tests)

Runtime end to end: lifecycle, errors, fail-closed policy, the watchdog, refusals.

Class	Test	What it pins	Cites
RuntimeTests	test_lifecycle_and_digest	Lifecycle and digest.	
RuntimeTests	test_hostile_value_is_contained_in_the_digest	Hostile value is contained in the digest.	

Class	Test	What it pins	Cites
RuntimeTests	test_errors_are_bounded_and_streaks_suppressed	Errors are bounded and streaks suppressed.	
RuntimeTests	test_error_streak_clears_on_success	Error streak clears on success.	
RuntimeTests	test_policy_violation_fails_closed_even_when_swallowed	Policy violation fails closed even when swallowed.	
RuntimeTests	test_undeclared_event_type_is_refused	Undeclared event type is refused.	
RuntimeTests	test_float_payload_is_refused	Float payload is refused.	
RuntimeTests	test_test_mode_refused_under_systemd	Test mode refused under systemd.	
RuntimeTests	test_impure_code_never_starts	Impure code never starts.	
RuntimeTests	test_unsafe_output_directory_is_refused	Unsafe output directory is refused.	
RuntimeTests	test_foreign_files_are_reported_not_touched	Foreign files are reported not touched.	
RuntimeTests	test_corrupt_ledger_refuses_and_changes_nothing	Corrupt ledger refuses and changes nothing.	
RuntimeTests	test_watchdog_pings_stop_while_an_observation_hangs	No background pinger: a blocked sense() starves the watchdog, so systemd would restart it.	
RuntimeTests	test_jitter_bound_is_zero_in_test_mode	Jitter bound is zero in test mode.	
RuntimeTests	test_jitter_bound_is_a_fraction_of_the_interval	Jitter bound is a fraction of the interval.	
RuntimeTests	test_jitter_bound_is_capped_for_long_intervals	86400s (the manifest maximum) times 5% would be 4320s; it must not drift by hours.	
RuntimeTests	test_jitter_disabled_in_test_mode_and_recorded_for_a_real_run	Jitter disabled in test mode and recorded for a real run.	
RuntimeTests	test_example_daemon_runs	Example daemon runs.	

Part 2: Adjudication of the r4.2 to r4.4 improvements

- **Asked:** the owner asked to "review and adjudicate these improvements": everything proposed in the combined review package (REVIEW-PACKAGE-COMBINED.md). That means the amendments to the four settled rulings and the open decisions in Packages A, B and C.
- **Authority:** adjudicated by Claude at the owner's request on 2026-09-30.
 - The owner's Class C authority is unchanged: each verdict stands unless the owner overturns it.
 - Two items interpret the owner's *own* rulings. They are marked **OWNER** and take effect only when the owner confirms them.
 - Nothing here activates a daemon, unlocks a class or changes the LTC.
- **Method:** each item was reviewed again with fresh eyes before a verdict, including the recommendations this repository made itself. Four of them are changed below (§3). Two findings were turned from "traced" or "not yet seen" into reproduced defects and fixed with tests (HF-34, HF-35). Every verdict names its evidence.
- **Verdict vocabulary:** ADOPTED, ADOPTED WITH MODIFICATION, DEFERRED (with a trigger), REJECTED. BUILT means done in this revision, with tests.

1. Summary

Group	Adopted	With modification	Deferred	Rejected	Built in r4.5
Settled-ruling amendments (5)	3 (one needs the OWNER)	1 (OWNER)	1	0	PD-70's clock basis (HF-34)
Package A: transport contract (5, plus the LTC markup)	4	2	0	0	—
Package B: Break Glass (5)	4	1	0	0	— (blocked on PD-72, PD-01.7, PD-40)
Package C (11 lines)	9	2	A-8, A-9, A-10 (inside one line)	0	DB-20 (HF-35, found during this review)
Seams N-1 to N-14	adopted as the required test list for their decisions				

Nothing is rejected: the second look changed four recommendations instead (PD-83, PD-85, PD-01.13 and the Phase 1 staleness checker), and tightened PD-76.

2. Two defects found and fixed while adjudicating

HF-34 (High): a backward wall-clock step brought HF-32 back. The r4.4 brief traced this; r4.5 reproduced it.

- **Reproduction:** a daemon blind on every cycle was killed and restarted every second against a 1.5 s limit, with its wall clock one hour behind after the first run.
 - On r4.4, all six restarts inherited 0 ms and SENSE_BLIND never fired.
 - Control, with the real clock: SENSE_BLIND on the second start.
 - Evidence: evidence/r4.5/clock_step_before.txt.
- **Fix (PD-70's amendment, adopted in §3.1):**
 - DAEMON_START, DAEMON_HEARTBEAT and DAEMON_STOP carry boot_id, boottime_ms (CLOCK_BOOTTIME) and blind_since_boottime_ms.
 - A restart within one boot measures inherited blindness on that clock.
 - Across a reboot the wall clock is used if it moved forward. Otherwise blind-at-the-limit is assumed, which leaves the one reacquisition cycle.
 - clock_basis is recorded in DAEMON_START and SENSE_BLIND.
 - A daemon event carries no boot time, so it is anchored at the preceding stamp. That can over-state blindness by at most $T/2$, in the safe direction.
- **Contract 3.1.0** (additive: no existing daemon can newly fail).
- **Tests** in tests/test_blind.py:
 - test_a_backward_clock_step_does_not_hide_blindness fails on r4.4.
 - test_start_heartbeat_and_stop_carry_the_boot_stamp fails on r4.4.
 - test_across_a_reboot_the_wall_clock_is_used_and_a_backward_one_assumes_the_worst.
 - test_an_event_after_the_last_stamp_is_anchored_at_that_stamp.
- After the fix, the reproduction fires SENSE_BLIND with the clock one hour behind (clock_step_after.txt).
- **Residual:** within a process the blind clock still runs on CLOCK_MONOTONIC, which does not count suspend. Across restarts it now counts suspend. A DGX server does not suspend, so this is recorded rather than changed.

HF-35 (High): the battery passed a daemon that never observes anything.

- **Found by** reviewing the battery for this adjudication: no check exercises the blind period. A daemon whose every cycle is unsettled records no error and no event.
- **Reproduction:** with its digest disabled, the fixture tests/fixtures/daemons/neverseeing got **RESULT: PASS** on r4.4 (evidence/r4.5/neverseeing_battery_before.txt). This is the same class of defect as HF-16 (r3), where the battery passed a daemon that failed every cycle.
- **Fix:** a new check, **DB-20 "observes within a blind limit"**.
 - It runs the daemon with a blind limit of five test intervals (1.0 s) over 12 cycles.
 - It requires accepted cycles in the heartbeats and no SENSE_BLIND.
 - It is a new ID, not an edit of DB-04, under PD-78's rule. DB-19 stays reserved for the direct memory reading (PD-76).
- **Tests:**
 - test_handoff.py::BatteryCatchesAlwaysFailingDaemonsTests::test_a_daemon_that_never_observes_fails_db20 fails on r4.4.
 - test_misc.py now pins the check list: 19 checks, ending DB-18, DB-20.
- **Result:** all four reference daemons pass DB-20 (10 accepted cycles each).

3. Verdicts

3.1 Amendments to settled rulings

Item	Verdict	Conditions, and why
PD-63: exit-code split (EXIT_POLICY $\neq$ SENSE_BLIND)	ADOPTED , built with contract 4.0.0	Keep SENSE_BLIND at 78 (the Observer's code). Choose EXIT_POLICY's new code from the Observer's exit table under PD-35, once that table is read, not before. Condition (seam N-12): one exit-code table generates the runtime constants, the unit's RestartPreventExitStatus, and the notifier's mapping; a test fails if any drifts
PD-63: where T is set	DEFERRED until PD-40 is built	No evidence of a problem today. Two places for one number is a cost worth paying only when per-host tuning is needed
PD-70: clock basis	ADOPTED and BUILT (HF-34, contract 3.1.0)	Reproduced, fixed, tested (§2)
PD-01.9: scope clarification	ADOPTED as the working interpretation; OWNER	It interprets the owner's own ruling, so it needs the owner's word. Until then this repository designs to it: the Blind Forester does not unlock critical; BF-5 is mandatory where a person could be harmed; it governs survival <i>mode</i> , never emergency <i>authority</i> ; PD-72 is a prerequisite
PD-76: mechanism wording	ADOPTED WITH MODIFICATION; OWNER (it rewords the owner's ruling)	Adopt "a direct cgroup peak: v2 memory.peak or v1 memory.max_usage_in_bytes". Modification: a v1 reading may decide PASS or FAIL only against a limit stated for v1 hosts, because v1 counts page cache differently. On the DGX (v2), only v2 readings are admissible. The battery's privileged step, creating a cgroup and moving the process into it, goes in a small separate helper with its own test, not inline (seam N-4)

3.2 Package A: the transport contract

Item	Verdict	Conditions, and why
PD-82 Bind, do not embed	ADOPTED	The LTC's own scope (LTC-01, "not a seventh pillar") and this repository's PD-52 agree
PD-83 Truncated reads raise	ADOPTED WITH MODIFICATION	Change from the recommendation: in contract 4.0.0, truncation <i>always</i> raises TooLarge. The opt-in "partial result" type is not built until a real daemon needs explicit partial semantics. No reference daemon does: dir-watch and git-watch both now fail the cycle. This removes seam N-10 instead of guarding it
PD-84 Register uses the LTC binding vocabulary	ADOPTED	A design rule for PD-40, with H02's three modifications
PD-85 Declared pacing	ADOPTED WITH MODIFICATION	Change from the recommendation: adopt the pacing field, but keep today's bounded random jitter as the default . Deterministic spread is available and recommended for new daemons. There is no measured problem with the current jitter on a host with four daemons: it is already zero in test mode, so it does not hurt testability. Changing every daemon's timing in a batch for a benefit nobody has measured is churn. Revisit when a host runs enough daemons to measure phase collisions
PD-86 Retry bounds hold over time	ADOPTED	Evidence: HF-28
LTC-H01 to H03, this repository's markup	ADOPTED for submission to the LTC reviewers	This repository cannot rule on the LTC. It forwards its markup (a total-time deadline for streams, digest-named trust evidence, bounds that hold over time)

3.3 Package B: Break Glass and operator-gated recovery

Item	Verdict	Conditions, and why
PD-01.10 Premise, scope, tiers	ADOPTED	The beacon tier (BG-S) comes after PD-01.3, PD-72 and PD-01.7. The acting tier (BG-A) is deferred with 3b to kernel v2
PD-01.11 Mechanisms	ADOPTED	A dormant relay; the owner's revised socket; single-use, epoch-bound envelopes
PD-01.12 Epochs, conservative restart rule	ADOPTED	Since HF-34, every record carries boot_id, which is the natural carrier for the epoch-per-boot rule (BG-9). Build the epoch on it
PD-01.13 Witness and flight recorder	ADOPTED WITH MODIFICATION	Change: where a person could be harmed (critical, still refused), the witness is separate hardware. At elevated, it is a separate process tree with its own clock source, and its independence is audited (U-7). "Separate process tree" alone is not enough evidence of independence at critical
PD-01.14 Recovery boundary	ADOPTED	It is also the Blind Forester's only exit, which completes BF-6

3.4 Package C

Item	Verdict	Conditions, and why
BF-1 to BF-6	ADOPTED , BF-5 mandatory	Each follows from a recorded invariant
PD-72 Supervisor/worker split	ADOPTED as a precondition ; built with the first active class	BF-1, LTC-14 and Package B all rest on it
PD-01.1, PD-01.2 Classes; consequence field	ADOPTED	The consequence field goes in contract 4.0.0
PD-01.3 Sentinel	ADOPTED WITH MODIFICATION	Ship it with ISA-18.2's minimum (A-5): one SENSE_DEGRADED record per blind streak, a rate limit, a defined response. Without that, alerts turn into fatigue
PD-69 Limits checked against each other	ADOPTED	Includes time windows (PD-86). DB-15's budget table is Phase 1
PD-71 (remainder)	ADOPTED , with DB-19	—
PD-75 to PD-81 Numbering	ADOPTED: alias	—
C-1 Battery exit codes	ADOPTED : 0/3/4 in contract 4.0.0	Its meanings change, so it moves with every other exit-code change, once
INCOMPLETE in CI	ADOPTED	Require PASS where a direct reading is possible; INCOMPLETE elsewhere only with a stated environment reason

Item	Verdict	Conditions, and why
Phase 1	ADOPTED WITH MODIFICATION	Change (seam N-5): the diverse staleness checker detects <i>change</i> . It records the ledger's size and head hash, and checks on its own monotonic clock that they changed within T. It does not compare a file's age against the wall clock, which has the HF-34 weakness
PD-64 to PD-68; A-1 to A-12	ADOPTED (PD-64 to PD-68; A-1, A-2, A-4, A-5, A-7, A-11, A-12). ADOPTED WITH CONDITION A-6: a shelving window extends T only by a human or a pre-approved rule, bounded and recorded (S-3). A-3 comes with PD-01.2. DEFERRED: A-8 with PD-01.5; A-9 and A-10 with the first active class or a certification path	—

3.5 Seams N-1 to N-14

ADOPTED as required tests. Each seam's "what a reviewer should check" becomes a test that must exist when its decision is built. BATTERY-REVIEW-R4.5.md §5 lists them as planned tests T-N1 to T-N14. Seam N-10 is removed by PD-83's modification.

4. What this changes in the order of work

1. **Done in r4.5:** HF-34 (contract 3.1.0) and HF-35 (DB-20).
2. **Owner confirms:** PD-01.9's scope, and PD-76's wording.
3. **Phase 1**, with the change-detecting staleness checker.
4. **DB-19** (direct memory reading), with the separate cgroup helper and the CI rule.
5. **Contract 4.0.0:**
 - the exit-code split (PD-63);
 - battery codes 0/3/4 (C-1, PD-81);
 - truncation always raises (PD-83);
 - the pacing field (PD-85);
 - the consequence field (PD-01.2);
 - PD-35, PD-39 and PD-49;
 - a migration hint for schema 3.
6. **PD-40, PD-72, PD-01.7**, then Package B's beacon tier.

Part 3: Review of the SQLite three-ledger recommendations

- **Status:** REVIEW AND ADJUDICATION. Decisions PD-87 to PD-92 are adjudicated by Claude at the owner's request, subject to the owner (ADJUDICATION-R4.5.md sets out that authority). Nothing here changes code. Four probes were run on this host (evidence/r4.5/); every other statement is marked as reasoned.
- **Revision:** r4.5, 2026-09-30, by Claude.
- **Asked:** the owner asked for a review of recommendations "for hardening and improving the daemon pattern's code map":
 - an append-only SQLite event store split into three ledgers (ingestion and events; memory reconciliation; external metadata), for an offline runtime kernel watched by a Git-based observer daemon;
 - systemd.path watchers on the SQLite write-ahead log, instead of polling;
 - Unix-domain sockets for all IPC, with POSIX permissions;
 - an application-layer SHA-256 hash chain, with a schema, an ingestion protocol and a verification loop.

1. Verdict in brief

The direction is sound, and much of it matches rules this repository already enforces for its own ledgers:

- one writer per ledger;
- append-only records;
- a hash chain from a genesis of 64 zeros;
- verification by a passive observer.

Five points need changing before the design is safe. Three of them are shown by probes on this host:

#	Point	Evidence	Change
1	The proposed payload JSON column changes what is stored. A declared type of JSON gets NUMERIC affinity in SQLite, so the text 123 is stored as the integer 123, and 1.50 as the real 1.5. The bytes the verifier reads are not the bytes that were hashed	Measured (sqlite_probes.txt, SQLite 3.45.1)	Declare payload TEXT NOT NULL CHECK (json_valid(payload)) in a STRICT table, which kept 1.50 as text in the probe. Verify the stored bytes; never re-serialize
2	Plain concatenation is ambiguous. previous_hash + timestamp + event_type +	Measured (sqlite_probes.txt)	Hash the canonical (JCS) encoding of the whole record as one object, including a

#	Point	Evidence	Change
	canonical_json(payload): event type A_B with payload {"x":1}, and event type A_ with payload B{"x":1}, give the same input and the same hash		sequence number and the previous hash. That is what this repository's ledger does (canonical.py, ledger.py)
3	A read-only observer cannot read the ledger after the writer closes. While the writer is open, a confined reader (uid 65534, directory not writable) reads fine. When the last writer closes, SQLite removes -wal and -shm, and the reader fails with "attempt to write a readonly database". The observer loses sight of the ledger exactly when it matters most: after the kernel stopped or crashed	Measured (sqlite_readonly_wal_probe.txt)	The writer keeps the WAL files persistent (SQLITE_FCNTL_PERSIST_WAL), or the observer verifies a snapshot the writer exports, or reads with immutable=1 only on a copy. Solve this before any read-only observer is built (PD-91)
4	A hash chain alone does not stop someone who can write the file. "Any altered bit breaks the link" is true only for an attacker who does not recompute the rows after the edit. Whoever can write the database can rewrite the chain from the edit onward	Reasoned (and the r4 known limit in WBS 3.0)	Anchor the head hash somewhere the writer cannot rewrite: the observer's Git commits, pushed off-host (PD-58). Also add BEFORE UPDATE and BEFORE DELETE triggers that abort, against accidents, not attackers
5	The observer must not govern canonical memory. "The passive observer daemon reads this ledger to resolve conflicts and govern the canonical memory state", and it "halts memory reconciliation" on corruption. Both are actions. An observer that resolves conflicts is at least an Advisor (class 2a), and halting is class 3	Against recorded rules: PD-01 §3, S-6, and the Observer's "observation never authorizes action"	The observer verifies, reports and proposes. The kernel, or a gate, writes supersession records and decides to halt. The observer's corruption finding is an escalation record the kernel acts on (PD-87)

2. The three ledgers

Adopted with modification (PD-92): split by writer authority, and name each writer. Separating operational metadata from semantic memory is right. What makes a split safe is that each ledger has exactly one writer, and that no component can write a ledger it should only read.

Ledger	Proposed	Adjudicated
Ingestion and events	Written only by the active kernel	Agree. The kernel is the single writer. Commits use BEGIN IMMEDIATE; in WAL mode EXCLUSIVE behaves the same. Each commit uses synchronous=FULL, because NORMAL in WAL mode keeps the database consistent but can lose the last commits on power loss, and a ledger that loses acknowledged records is not a ledger
Memory reconciliation	Read by the observer, which "resolves conflicts and governs"	Modified. The single writer is the kernel or a governance gate. The observer reads it, verifies supersession paths and proposes resolutions (class 2a). Validity intervals follow S-8 and the HF-34 lesson: they state their clock basis, and order comes from sequence, never from wall-clock timestamps
External metadata	Written by WASM plugins "via its socket"	Agree, made explicit. A small ledger service owns the file and is its single writer. Plugins send requests over a socket and never hold the file, so the service, not the plugin, computes the chain

Row rules for all three, which are this repository's ledger rules applied to SQLite (PD-88):

- a STRICT table;
- seq INTEGER PRIMARY KEY, contiguous and checked by the verifier;
- record TEXT NOT NULL, holding the exact canonical bytes that were hashed;
- previous_hash and current_hash, each CHECK (length(x) = 64 AND x NOT GLOB '^[^0-9a-f]*');
- current_hash UNIQUE;
- triggers that abort any UPDATE or DELETE;
- genesis 64 zeros;
- timestamps are informational and never order anything.

3. Event-driven wake with systemd.path

Adopted with modification (PD-89): an event is a hint that shortens the wait; it is never the liveness signal.

- **A path unit starts a unit; it cannot wake a resident process.** If the observer is a long-running daemon (this pattern's shape), a path unit's activation does nothing while the service is already running. It suits a oneshot script, not a resident daemon.
- **Events are coalesced and can be dropped.** Writes that arrive while the triggered unit is still running are coalesced. Path units are rate-limited in recent systemd (TriggerLimitIntervalSec=, TriggerLimitBurst=): when the limit trips, the unit fails and stops triggering. inotify itself can overflow its queue. So completeness can never rest on events.
- **Silence and death look the same.** With push-only wake, "nothing was written" and "the writer is dead" produce the same silence. This pattern's blind period and heartbeat exist to tell those apart (PD-63, PD-70), and they need a periodic cycle.
- **"The exact millisecond" overstates it.** A write to -wal is not yet a commit, so a woken reader can see nothing new (reasoned). In this pattern that case is ctx.unsettled: nothing stable was read.

Design:

- The resident daemon keeps its poll interval as its **floor**: the most it can be late, and its liveness proof.
- A path unit starts a tiny oneshot that sends the daemon a wake signal, and the daemon's sleep ends early.
- A missed or coalesced event costs at most one interval.
- Nothing inside the skeleton needs inotify. That matters because ctypes is confined to `landlock.py` and `probes.py` (`tests/test_layers.py`).

Build it when a daemon needs latency below its interval. No reference daemon does today.

4. Unix-domain sockets for IPC

Adopted as execution-profile guidance, not a contract rule (PD-90).

- **It belongs to the execution profile, not the contract.** Under the transport contract, the IPC mechanism is a provider and execution-profile choice (LTC-02 forbids a contract that requires one primitive, and PD-82 binds the pattern to the LTC at its edges). "Use UDS" therefore goes in the execution profile, not in the LTC or the daemon contract.
- **Pathname sockets only.** Put them in a protected directory and check the peer with `SO_PEERCRED`, as Package B's socket does (BG-6). Avoid abstract-namespace sockets: they have no file permissions. Where they cannot be avoided, Landlock ABI 6 and later can scope them.
- **POSIX permissions cannot separate plugins inside one process.** "A WASM plugin has write access to the metadata ledger via its socket but is physically denied the reconciliation ledger" holds only if each plugin runs as its own process with its own uid. Plugins hosted inside one runtime process share its uid and its open sockets. There the separation has to come from the WASM host's capability handles (WASI), not from file permissions (reasoned).
- **The latency claim is overstated.** UDS avoids the TCP/IP stack and is faster for small messages, but for bulk ingestion the difference is modest. The strong argument for UDS is security: no network exposure, kernel-checked peer identity, and Landlock can deny TCP entirely (DB-17 shows it does for daemons today).

5. What changes in the daemon pattern's code map

Recommendation	Where it would land	When
Hash-chain rules shared by every Spark ledger (PD-88)	<code>canonical.py</code> stays the reference encoding; the planned independent verifier (DB-22) is written to verify both this JSONL format and a SQLite table built to PD-88	With DB-22
A read-only SQLite capability for daemons (PD-91)	<code>context.py</code> gains <code>ctx.sqlite_rows(db, sql, params, max_rows): SELECT only, mode=ro URI, PRAGMA query_only, a progress-handler time limit, a row cap that raises TooLarge (PD-83), and bounded read transactions (a long read blocks checkpoints and grows the WAL). <code>manifest.py</code> declares the database paths. <code>landlock.py</code> needs read access to the database and its <code>-wal</code> and <code>-shm</code></code>	Deferred until the kernel's ledger exists and point 3 is solved
A reference observer for a SQLite chain	A fifth reference daemon, <code>sqlite-ledger-watch</code> , class <code>la</code> : it verifies, records the head and anchors it, and escalates on corruption. It never halts anything itself	With PD-91
Wake hints (PD-89)	<code>runtime.py</code> : a signal ends the sleep early; <code>unitgen.py</code> : an optional path unit and oneshot; <code>manifest.py</code> : <code>trigger.wake_on</code> paths, with the interval required as the floor	When a daemon needs it
This pattern's own ledgers	No change. They stay JSONL with one fsync per record (PD-02, PD-56). Two formats are acceptable if one independent verifier checks both	—

New seams, which join the reviewer-focus list:

- **N-15**: the observer, the WAL files and Landlock (point 3).
- **N-16**: the ledger service and the plugin sockets (single writer).
- **N-17**: the wake hint and the poll floor, where an event must never replace the liveness cycle.
- **N-18**: a head anchor that the kernel cannot rewrite.

6. Decisions

PD-87. Observers verify and propose; they never govern. The reconciliation ledger's writer is the kernel or a gate. The observer's corruption finding is an escalation the kernel acts on. Verdict: **ADOPTED** (design rule for the kernel's memory store).

PD-88. One set of hash-chain rules for every Spark ledger, SQLite included (§2): canonical encoding of the whole record, stored bytes verified as stored, a `STRICT` schema, abort triggers, `synchronous=FULL`, sequence-ordered, head anchored off the writer's reach. Verdict: **ADOPTED**.

PD-89. Event wake is a hint; the interval is the floor (§3). Verdict: **ADOPTED WITH MODIFICATION** (hint via signal, not path-unit activation of a resident daemon). Build when needed.

PD-90. UDS in execution profiles, with peer credentials; per-plugin separation needs per-plugin processes or capability handles (§4). Verdict: **ADOPTED as profile guidance**.

PD-91. A read-only SQLite capability and a reference observer. Verdict: **DEFERRED**. Triggers: the kernel's ledger exists, and point 3 is solved and probed under Landlock on the DGX's SQLite version.

PD-92. The three-ledger split, by writer authority, with each writer named (§2). Verdict: **ADOPTED WITH MODIFICATION**.